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<?php
/* 3 a. Develop a PHP Program(s) to check given number is:
    (i) Odd or even
    (ii) Divisible by a given number (N)
    (iii) Square of a another number

    3a_odd_divisible_square.php
*/
$num = 9;    // given number

// check given number is (i) Odd or even
if ( $num % 2 == 0 ) //Modulo arithmetic operator, remainder of $num / 2
    echo "<br /> $num is even";           // Operator Precedence,
else                                     // % higher than ==
    echo "<br /> $num is odd"; //or for clarity, ( ( $num % 2 ) == 0 )

// check given number is (ii) Divisible by a given number (N)
//      solution based on (i)
$n = 5;                                     // a divides b, written as a | b
if ( $num % $n == 0 ) // there exists  $\exists$  integer k, such that  $b = a * k$ 
    echo "<br /> $n divides $num"; // or  $b \bmod a = 0$  , b divided by a
else                                     // leaves no remainder
    echo "<br /> $n does not divide $num"; // another way, using loop

// check given number is (iii) Square of a another number
$another_num = 3;    // is square of a another number = given number
if ( ( pow($another_num , 2) ) == $num ) // pow ( number, exponent )
    echo "<br /> $another_num squared is $num";    // $ php --rf pow
else
    echo "<br /> $another_num squared is not $num";

?>

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For more on , please check ,
Arithmetic Operators
https://www.php.net/manual/en/language.operators.arithmetic.php
Operator Precedence
https://www.php.net/manual/en/language.operators.precedence.php
Control Structures
https://www.php.net/manual/en/language.control-structures.php
Math Functions
https://www.php.net/manual/en/ref.math.php
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